

POAMS: LECTURE 11
QUANTUM MECHANICS AND THE DISCRETE NATURE OF
LIGHT-ENERGY

Having discussed the quantisation of all physical measures and the origins of Quantum Mechanics in the previous lecture, I shall now continue with my potted history of Quantum Mechanics, leading to the second stage in its development to its current form. Along the way, I shall revisit the EPR controversy, first discussed in Lecture 6. I shall then continue with a study of the discrete nature of light-energy. This lecture is based on some of 'Light Speed...' Chapter 5.

1. The quantisation of physical measures

To summarise a vital point from the previous lecture, from the standpoint of Normal Realism, since all fundamental physical quantities are based on *observations*, it follows that on a small enough scale, they must be ultimately *discrete*, rather than continuous as in classical physics. Although all physical measures are ultimately discrete on a small enough scale, the fact that these scales are so small makes physical measures appear, for all practical purposes, to be continuous on the macro-scale. This is known as the *correspondence principle* in Quantum Mechanics.

As in the previous lecture, let me expand on this idea of the discreteness of physical measures further. From the holistic standpoint of Normal Realism, all fundamental measures of physical phenomena are interrelated. Hence, whenever a practical reduction in any arbitrary unit of such a measure is contemplated, we have to think of it as being carried out within the context of the whole relational network of such measures. It seems reasonable, therefore, to suppose that for each particular phenomenon, at some stage in the reductive analysis, the units of certain fundamental measures are optimal in the sense that further subdivisions of a particular unit will cause others to increase. The minimum values of these units of fundamental physical measures, with respect to the whole relational system, can be thought of as the natural, indivisible units of such measures.

Our Normal Realist standpoint is supported by the first stage in the development of Quantum Mechanics. This began with Planck's introduction of the quantisation of energy, in order to explain the Rayleigh-Jeans results concerning *black body* radiation, and lasted until about 1925.

Following Planck's pioneering work and Einstein's proposals concerning the *photoelectric effect*, it was soon discovered that a *discrete structure* was persistent throughout *all physical phenomena*. Any physical property is *quantised*, the **quantum** being the minimum amount of any physical property involved in an interaction. [*End of recap*]

2. The EPR controversy revisited

Recall that, from Lecture 6, the *Copenhagen interpretation* of Quantum Mechanics was developed initially by Bohr in the early 1920's. This was part of the first stage in the development of Quantum Mechanics. In this interpretation, a quantum 'particle' does not exist in one state or another, but in all its possible states at the same time. It is only when we observe its state that a quantum 'particle' is essentially forced to choose one probability, and that's the state we observe. This interpretation does not allow any separation between observer and the system observed, regardless of distance. This implies the existence of some type of *instantaneous action-at-a-distance*, as discussed in Lecture 6.

The 1935 paper, written by Einstein in collaboration with Podolski and Rosen challenged Bohr and his followers on the grounds of Einstein's assumption of *locality*, according to which it is nonsense to expect that determining the state of a particle at one place is tantamount to determining, immediately, the state of another particle somewhere else, regardless of distance. Such instantaneous action-at-a-distance is forbidden by Special Relativity, according to which 'nothing can travel faster than the speed of light'. Einstein and his co-workers argued that, at the very least, Bohr's assertion of non-locality must be incomplete in some sense. More specifically, the 1935 paper provides a *thought experiment*, which maintains that the description of physical reality provided by Quantum Mechanics is incomplete – it is speculated that it should be possible to construct a complete theory containing additional local variables (so-called *hidden variables*), an idea known as *local realism*.

This clash between Bohr and Einstein continued unresolved for almost thirty years and became known as the 'Einstein-Podolski-Rosen (EPR) controversy'. Then in 1964, John Bell of CERN suggested a way of deciding, on experimental grounds, whether or not Bohr's instantaneous

action-at-a-distance existed. Implicit in Bell's work is the proposition that the determinism of classical physics is fundamentally incapable of describing Quantum Mechanics. Bell provided the conceptual foundation of the *Bell inequality experiments*, which are designed to test Quantum Mechanics against Einstein's concept of local realism. If a Bell inequality is violated, then local realism cannot hold.

To date, all experiments, such as those carried out by John Clauser in 1978 and Alain Aspect in 1982, have found that any Bell inequality is violated, so that Einstein's hypothesis of additional local variables is inconsistent with the way physical systems are observed to behave. If a Bell inequality were true, then it would be possible to rig two separated objects to correlate their properties more than 75% of the time without the existence of any instantaneous action-at-a-distance.

3. De Broglie waves

Recall that, in order to explain the results of the photoelectric effect, Einstein suggested that electromagnetic radiation actually consists of *action* quanta, h , which he called *photons*, each of energy given by the Planck relation

$$E = h\nu \quad (1)$$

This appeared to be confirmation, by modern physics, of Newton's once prestigious *particle theory* of light. In this way, Einstein created an apparent paradox. How could light have both wave-like characteristics and particle characteristics? The answer to this dilemma, first proposed by Einstein and Louis De Broglie (1892-1987), was purely *ad hoc*, namely, that light has wave characteristics as well as particle characteristics. And so, counter to all commonsense, the photon was conceived to be a *wave-particle*!

In 1924, De Broglie extended this wave-particle concept of light to the motion of particles of any sort. He associated a theoretical wave with any particle, which could be used to represent properties of that particle that conform to equations that describe waves. These *De Broglie waves* were used to provide a new model of the atom, in which standing waves are associated with electrons. These waves form harmonic patterns around the

nucleus of an atom, at distances from the centre where the beginning and end of the wave-cycle are perfectly in phase.

4. The second stage in the development of Quantum Mechanics

The first stage in the development of Quantum Mechanics consisted of a mixture of both classical and non-classical concepts and was not completely satisfactory. For example, it led to the wave-particle paradox involving light already described. The second stage of development was achieved by Erwin Schrödinger (1887-1961) and Werner Heisenberg (1901-1976) and had its beginnings during 1925. De Broglie's concept of the underlying standing-wave in an atom was further developed by Schrödinger early in 1926, by the introduction of a *wave equation* that satisfactorily described all quantum aspects of physical systems. Later that same year, Max Born (1882-1970) subjected Schrödinger's wave function to a probabilistic interpretation, describing it not as a continuous wave but as a pure *probability amplitude*. In this interpretation, exactness of position and motion of elementary particles disappears at the quantum level. All that is left are pure wave-like 'probability functions'. The traditional notion of a particle as a hard material entity was thus fast disappearing. It was to be replaced by the concept of a real physical state brought about by the 'collapsing' of an underlying 'probability wave-function', by the mere fact of observing it.

In 1927, Heisenberg proposed his *Uncertainty Principle*, according to which observing the state of an elementary particle requires an energy-interaction with that particle, which alters its state. This means that so far as observation is concerned, the existence and state of particles on the ultimate quantum level is impossible to determine, since observing a particle inevitably alters whatever might be thought of as its 'unobserved' state. Because of this, Heisenberg concluded that so-called 'elementary particles' cannot exist at the ultimate micro-physical level. Due to these developments, what is now known as Quantum Mechanics slid further and further into the mystery we now associate with it.

From these ideas, the Schrödinger-Heisenberg-Born model of the atom was developed to replace the earlier Bohr model. In this model, a wave equation specifies three quantum numbers for the wave function of an electron. A *spin* quantum number was added at a later date. In this atom, the

Uncertainty Principle and the de Broglie hypothesis are incorporated and as such it is only possible to deal with the probability of finding an electron in phase space, discarding the ideas of continuous circular or elliptical orbits as in the Bohr model.

From our Normal Realist viewpoint, what concerns us here is that despite the undoubted practical efficacy of Quantum Mechanics and its later developments, it is based, like 'light travelling *in vacuo*', on concepts of things and processes that are altogether unseen and un-sensed, things existing in realms apart from ordinary (observable) reality. Moreover, these 'hidden mechanisms' are often conceived by analogy with our experiences on the macro-scale. However, these analogies are as inappropriate for understanding processes at the quantum level as it would be, at the level of information theory, to try and analyse a video-screen picture of a horse in terms of a knowledge of nothing but horses.

As an alternative to the Schrödinger-Heisenberg-Born atom, we shall provide our Normal Realist approach to the atom, in the form of the POAMS model, in a later lecture, once POAMS has been fully formulated.

5. The discrete nature of light-energy

As discussed previously, in our Normal Realist account of Relativity, light is to be thought of as *observational information*, rather than consisting of 'particles' or 'waves', 'travelling at speed c '. Stripped of all its mechanistic precepts, light quanta are simply informational bits, in bar-code-like arrangements, out of which, ultimately, our knowledge of the world is constructed. What we observe is therefore a hologram-like presentation.

According to the Planck relation, equation (1), light (in the sense of *all electromagnetic radiation*) is emitted and absorbed in discrete and irreducible units of Planck's constant, h . Hence, if we wish to think of light as 'travelling at speed c ', then we have to think of these quantum *bits*, conventionally called *photons*, as all travelling at the same speed. But then, according to this interpretation, these photons have to possess mass, and Special Relativity has it, that at that 'speed', the observational mass of any travelling 'particle' becomes infinite (see equation (7), Lecture 9). This is yet another argument against the interpretation that light 'travels' in the

usual sense. In contrast, according to our Normal Realist philosophy, light, as energetic pixels of observational information, does not travel in this way but is simply a transfer of *disembodied energy* from one place to another within the phenomenal framework.

We thus think of a 'photon' as a quantum of transferable energy from one observational location to another. This means that the Greek-based word 'photon', which Einstein used to describe the light quantum, is inappropriate for our purposes due to its unwanted 'space travelling particle' connotations. For this reason, we prefer to use the Latinised version, *photum* (plural *phota*) to describe the Planck quantum insofar as it carries no mechanical overtones of the usual ballistic kind. This dispenses with the apparent conflict of the 'wave' versus 'particle' theory of light and the consequent notion of light as a 'wave-particle'. It is to be noted that apart from its lack of the usual separatist, space-travelling connotations, the photum is the same in all other respects as the photon.

As discussed in Lecture 6, far from having a 'speed', any incorporeal light interconnection is inherently proper time instantaneous. In this sense, energy transfer in light interaction is of a *potential* rather than a *kinetic* kind. Indeed, it is known as *quantum potential* in some quarters. This purely potential nature of quantum interconnection is borne out by the well-known Thomas Young (1773-1829) two-slit interference experiment, first performed in 1801. In this experiment, a light source is shone through a screen with two slits. This produces an interference pattern of alternating light and dark fringes on a plate behind the screen. This pattern cannot be explained if light has particle-like characteristics, which led Young to deduce that in the experiment light 'travels' in waves, and so thus supported the 'wave theory' of light. Interpreted as photons, the light quanta seem to 'know', ahead of time, how to avoid those areas on the plate where their frequencies, if they 'landed', would be out-of-phase with one another and 'steer', instead, in correspondingly greater numbers towards those areas where they would be in-phase. This is in such a way as to conserve (as through by telepathy from the 'travelling photon' point of view) the overall spread of illumination at the distant end. The mystery of the results of Young's experiment disappears when explained in terms of proper time instantaneous potential energy transfer, which does not induce us to think of light as travelling either in the form of waves or particles.

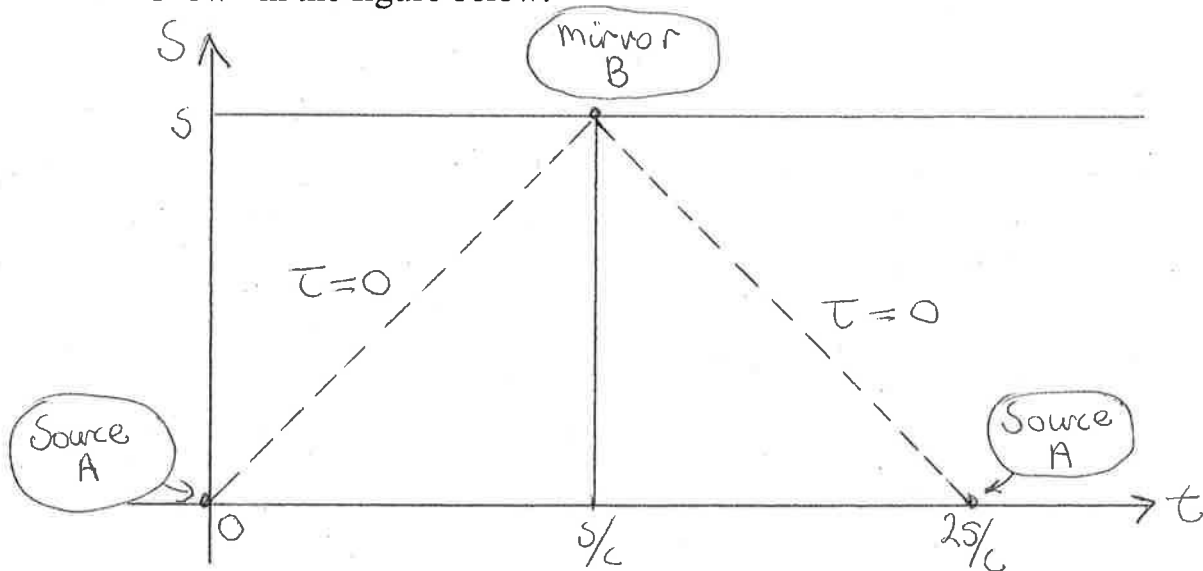
Something else which is certain, from this point of view, is that the light-energy cannot 'set out' on some course in the vague hope of sometime in the future meeting either an absorber or a 'barrier' of some sort. Conservation of energy demands that every quantum of interaction has to be an instantly consummated, double-ended affair, a one-ended quantum jump being impossible to conceive as a one-ended piece of wire. Our proposal, therefore, is to think of the atoms involved in the energy transfer as being as being in direct and reciprocal resonance. This is in accordance with Newton's Third Law of equal and opposite action and reaction in what has been called *quantum touching* (as mentioned in Lecture 6).

In our Normal Realist approach then, the photum (photon) is treated as a single irreducible quantum event common to both an 'object' and the observer. There is no proper time separating whatever we may think of as the 'beginning and end' of a quantum interaction. Space and time, as we normally perceive them, therefore, cannot exist on that quantum level, any more than a video drama can exist at the individual screen pixels. These dimensions of physical phenomena exist solely on the macro-phenomenal level, optically projected from informational patterns and sequences of these quantum 'pixels'. On the quantum-phenomenal level, the atoms of bodies interact by direct resonance, with nothing existing or happening in between.

For example, if on the macro-level, these resonances are between the atoms of a distant galaxy and those in the eye of an astronomer or a photographic plate, then the distance of that object is an optical projection constructed from information based on statistical numbers of those immediately perceived, intrinsically distance-less quantum events. This is in the same way that in some ultra-fine-grained pointillist painting, the length, depth and breadth of a landscape are optically projected from whole arrangements of paint-spots in a way which would be impossible from examining those paint-spots themselves.

In these purely information-sequential terms then, how do we account for the observational delay in two-way signalling, as, for example, in Michelson's experiments to determine the 'speed of light'? Our answer is as follows.

A single photum has no intrinsic spatial or temporal extension, so in itself, it cannot signify any measure of observational distance. Rather, the distance, s , of a source relative to an observer O , is the manifestation of distributions of large numbers of photas, of which that particular atomic source is just one among many, all observed more or less together at O . Since all distances in observer-space are also times in the ratio of units c , the emitter and absorber of a photum, which are non-separated in space and (proper) time, are separated in observer time by the observational time interval s/c . Thus, if A is a light-source and B is a mirror situated a distance s from A , a photum originating at A , although seen instantly at B by observer O , with respect to the proper time of the photum itself, cannot reach B in O 's extrapolated observer time until the lapse of time s/c , as shown in the figure below.



The loss of the photum at A and the gain of that same photum by B affects both A and B simultaneously. However, since B cannot both gain and lose a quantum of energy in the same instance, the response of a mirror at B , *i.e.*, its 'reflection', has to be another photum contact in which, in this instance, a quantum of energy is lost by B and gained (again proper time simultaneously) by A . This involves another lapse of observational time s/c . Hence, the overall observational time recorded at A , for the signal transaction will be $2s/c$, as shown in the figure. Hence, what we customarily call the 'transmission and reflection of the photon' is no more than the 'cinematographic' extrapolation, in A 's observation of the two, separate, intrinsically instantaneous blips illustrated in the figure. We may

summarise this situation by saying that each *one-way* leg of photum interaction is instantaneous, whilst all *two-way* photum interactions are time-retarded.

This analysis shows that the instantaneous to-and-fro linkages between photons in the sequences of which macro-phenomenal time ultimately analyses, cannot be *transitive*. That is, although both the photonic interaction between *A* and *B* and the photonic interaction between *B* and *A* upon reflection by the mirror are both proper time instantaneous, this does not imply that the photonic interaction between *A* and itself upon reflection is also proper time instantaneous. Hence, there cannot be, at least not at the final microphysical level, any universal, or non-local, *single present-moment* in which all these instantaneous connections occur together. If there were, it would be a case of 'cosmic gridlock'.

In summary, the energy transfer in light interaction is proper time instantaneous in accordance, not only with the time-dilation formula, but also with conservation of energy. The time-delay which is usually associated with these phenomena is not a feature of the quanta themselves, but only of their time-sequences in macroscopic observation.

In the next lecture, I will continue with the current theme and we will study causality at the quantum level, photum interaction and conservation of four-momentum, before moving on to our alternative, Normal Realist, derivation of the Balmer-Rydberg formula, first studied in the previous Lecture.

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